



Sreeja T 251801325-2025-AIDS ▾

S2**Started on** Wednesday, 7 October 2026, 9:28 PM**State** Finished**Completed on** Wednesday, 7 October 2026, 9:30 PM**Time taken** 1 min 33 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int arr[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &arr[i]);
11     }
12
13     for (int i = 0; i < n; i++) {
14         for (int j = i + 1; j < n; j++) {
15             if (arr[i] == arr[j]) {
16                 printf("%d", arr[i]);
17                 return 0;
18             }
19         }
20     }
21
22     return 0;
23 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Sreeja T 251801325-2025-AIDS ▾

S2**Started on** Wednesday, 7 October 2026, 9:30 PM**State** Finished**Completed on** Wednesday, 7 October 2026, 9:37 PM**Time taken** 6 mins 36 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int arr[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &arr[i]);
11     }
12     int slow = arr[0];
13     int fast = arr[0];
14
15     do {
16         slow = arr[slow];
17         fast = arr[arr[fast]];
18     } while (slow != fast);
19     slow = arr[0];
20
21     while (slow != fast) {
22         slow = arr[slow];
23         fast = arr[fast];
24     }
25
26     printf("%d", slow);
27
28     return 0;
29 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Sreeja T 251801325-2025-AIDS ▾

S2**Started on** Wednesday, 7 October 2026, 9:37 PM**State** Finished**Completed on** Wednesday, 7 October 2026, 9:40 PM**Time taken** 2 mins 19 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int T;
5      scanf("%d", &T);
6
7      while (T--) {
8          int m, n;
9
10         scanf("%d", &m);
11
12         int a[m];
13         for (int i = 0; i < m; i++) {
14             scanf("%d", &a[i]);
15         }
16
17         scanf("%d", &n);
18
19         int b[n];
20         for (int i = 0; i < n; i++) {
21             scanf("%d", &b[i]);
22         }

```

```
23
24     int first = 1;
25
26     for (int i = 0; i < m; i++) {
27         for (int j = 0; j < n; j++) {
28
29             if (a[i] == b[j]) {
30                 if (!first)
31                     printf(" ");
32
33                 printf("%d", a[i]);
34                 first = 0;
35                 break;
36             }
37
38             if (b[j] > a[i]) {
39                 break;
40             }
41         }
42     }
43
44     printf("\n");
45 }
46
47 return 0;
48 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



[Back to Course](#)



Sreeja T 251801325-2025-AIDS ▾

S2**Started on** Wednesday, 7 October 2026, 9:40 PM**State** Finished**Completed on** Wednesday, 7 October 2026, 9:42 PM**Time taken** 1 min 54 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int m, n;
9
10        scanf("%d", &m);
11        int a[m];
12
13        for (int i = 0; i < m; i++) {
14            scanf("%d", &a[i]);
15        }
16
17        scanf("%d", &n);
18        int b[n];
19
20        for (int i = 0; i < n; i++) {
21            scanf("%d", &b[i]);
22        }
23    }

```

```
23
24     int i = 0, j = 0;
25     int first = 1;
26
27     while (i < m && j < n) {
28         if (a[i] == b[j]) {
29             if (!first)
30                 printf(" ");
31
32             printf("%d", a[i]);
33             first = 0;
34
35             i++;
36             j++;
37         }
38         else if (a[i] < b[j]) {
39             i++;
40         }
41         else {
42             j++;
43         }
44     }
45
46     printf("\n");
47 }
48
49 return 0;
50 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.



[Back to Course](#)



Sreeja T 251801325-2025-AIDS ▾

S2**Started on** Wednesday, 7 October 2026, 9:42 PM**State** Finished**Completed on** Wednesday, 7 October 2026, 9:49 PM**Time taken** 6 mins 40 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1 | #include <stdio.h>
2 |
3 | int main() {
4 |     int n;
5 |     scanf("%d", &n);
6 |
7 |     int A[n];
8 |
9 |     for (int i = 0; i < n; i++) {
10 |         scanf("%d", &A[i]);
11 |     }
12 |
13 |     int k;
14 |     scanf("%d", &k);
15 |
16 |     for (int i = 0; i < n; i++) {
17 |         for (int j = 0; j < n; j++) {
18 |
19 |             if (i != j && A[j] - A[i] == k) {
20 |                 printf("1");
21 |                 return 0;
22 |             }
23 |         }
24 |     }
25 |
26 |     printf("0");
27 |
28 |     return 0;
29 | }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Sreeja T 251801325-2025-AIDS ▾

S2**Started on** Wednesday, 7 October 2026, 9:50 PM**State** Finished**Completed on** Wednesday, 7 October 2026, 9:51 PM**Time taken** 50 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int A[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &A[i]);
11     }
12
13     int k;
14     scanf("%d", &k);
15
16     int i = 0, j = 1;
17
18     while (i < n && j < n) {
19         if (i == j) {
20             j++;
21         }
22         else {
23             int diff = A[j] - A[i];
24
25             if (diff == k) {
26                 printf("1");
27                 return 0;
28             }
29             else if (diff < k) {
30                 j++;
31             }
32             else {
33                 i++;
34             }
35         }
36     }
37
38     printf("0");
39

```

```
40 |         return 0;  
41 |     }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

